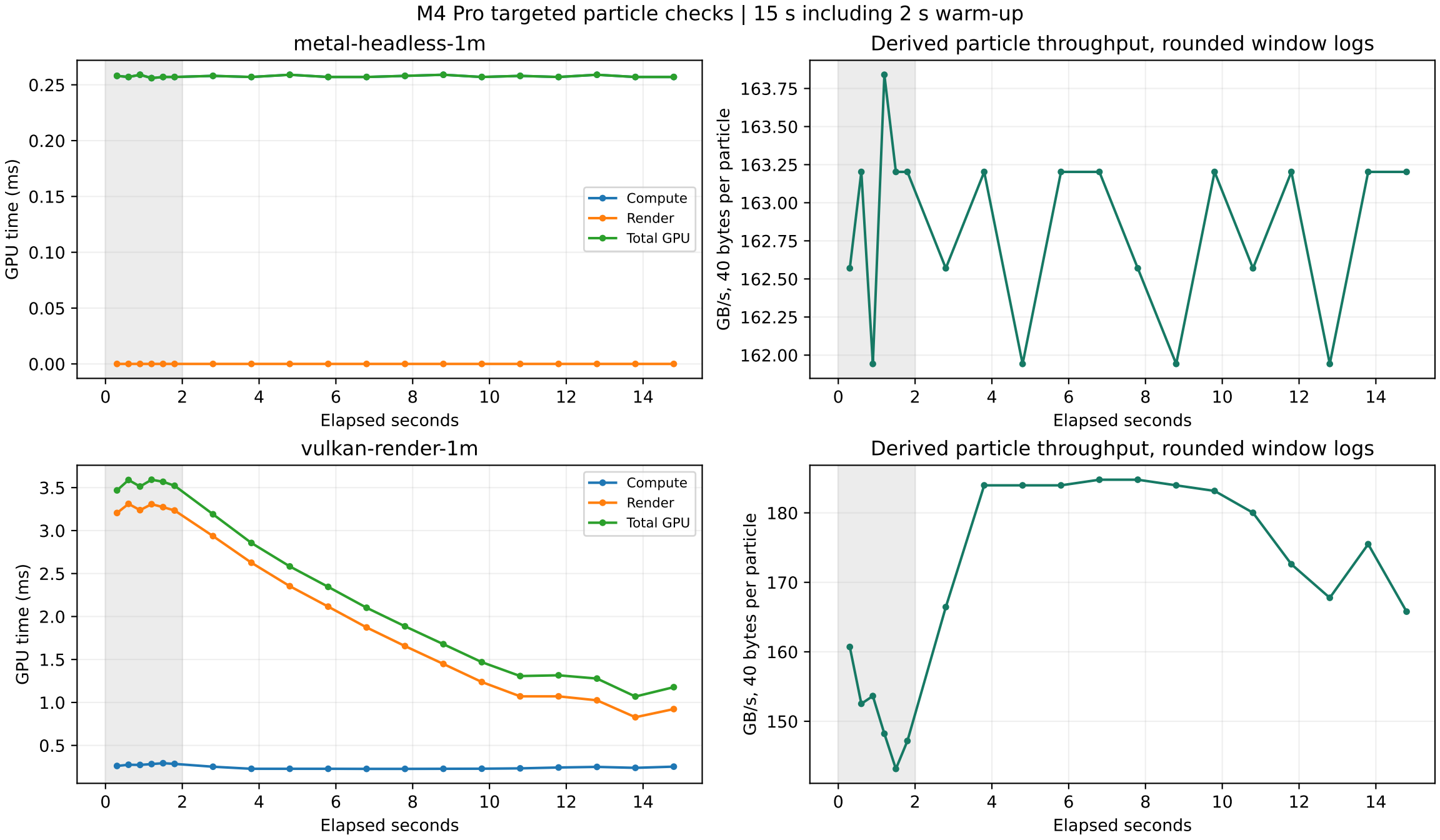
M4 Pro targeted particle checks | 15 s including 2 s warm-up
metal-headless-1m
GPU time (ms)
Elapsed seconds
Compute
Render
Total GPU
Derived particle throughput, rounded window logs
GB/s, 40 bytes per particle
Elapsed seconds
vulkan-render-1m
GPU time (ms)
Elapsed seconds
Compute
Render
Total GPU
Derived particle throughput, rounded window logs
GB/s, 40 bytes per particle
Elapsed seconds